

ApexPlanet Data Analytics Internship

Task 3 — Deep-Dive Analysis & Interactive Dashboard

Project: Sales Performance Deep Dive

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1. Executive Summary

This deep-dive extends the Task 2 exploratory analysis into a management-oriented sales performance view. The analysis focuses on product performance, geographic performance, customer value, demographic revenue patterns, and the relationships of Quantity and Unit Price with Total Sales. The objective is to identify where revenue is concentrated and provide an interactive dashboard structure for further exploration.

The cleaned dataset contains 1,000 records, 992 unique orders and 947 unique customers. Total revenue is ■139,399,439.65, total quantity is 5,435 units, and average order value is ■140,523.63.

2. Core KPIs

KPI	Formula	Result
Total Revenue	SUM(Total_Sales)	■139,399,439.65
Total Orders	DISTINCTCOUNT(Order_ID)	992
Total Customers	DISTINCTCOUNT(Customer_ID)	947
Total Quantity	SUM(Quantity)	5,435
Average Order Value	Total Revenue / Total Orders	■140,523.63

3. Deep-Dive Business Question

Which products, locations, customer groups and demographic segments contribute most to overall sales revenue, and which measurable variables are most closely associated with Total Sales?

4. Product Performance

Product	Revenue	Quantity	Average Revenue
Laptop	■25,443,008.51	970	■149,664.76
Mobile	■25,335,573.19	1,008	■137,693.33
Book	■25,031,689.40	977	■140,627.47
Rice	■22,231,711.28	826	■145,305.30
Chair	■21,521,561.48	855	■135,355.73
Shoes	■19,835,895.79	799	■127,153.18

Finding: Laptop has the highest product revenue at ■25.44M, while Mobile has the highest quantity sold at 1,008 units. This distinction is useful because the product with the highest revenue is not the same as the product with the highest volume.

5. Geographic Performance

City	Revenue
Patna	■19,285,966.89

Kolkata	■18,884,349.57
Bengaluru	■18,773,574.32
Mumbai	■18,757,050.17
Hyderabad	■17,166,766.87
Delhi	■16,097,079.00
Pune	■14,513,175.90
Gaya	■14,380,859.39
Unknown	■1,540,617.54

Finding: Patna records the highest city revenue at ■19.29M. The dashboard should allow users to select a city and cross-filter product, customer and demographic visuals.

6. Demographic Revenue Analysis

Age Group	Revenue
18–24	■21,225,484.12
25–34	■27,514,867.44
35–44	■32,800,713.26
45–54	■27,324,086.22
55–65	■30,534,288.61

Finding: The 35–44 age group generates the highest revenue at ■32.80M. However, the correlation between individual Age and Total Sales is approximately 0.001, so age should be treated as a segmentation variable rather than a linear sales driver.

7. Customer Value Analysis

Customer ID	Customer Name	Total Spending
CUST2062	Customer_254	■493,677.50
CUST4706	Customer_200	■492,174.10
CUST1711	Customer_36	■490,866.40
CUST4869	Customer_392	■485,667.80
CUST7416	Customer_191	■482,551.90

The top five customers each generated roughly ■482.6K–■493.7K in recorded spending. For the interactive dashboard, customer-level revenue should be used to support drill-down and a rule-based customer segmentation view.

8. Relationship Analysis

Variable Pair	Correlation (r)	Interpretation
Age vs Total Sales	0.001	Virtually no linear relationship
Quantity vs Total Sales	0.647	Moderate positive association
Unit Price vs Total Sales	0.686	Moderately strong positive association

These are associations, not causal effects. The stronger positive association of Unit Price and Quantity with Total Sales supports using both variables in the dashboard as revenue-driver views.

9. Customer Segmentation Framework

A practical business-rule segmentation can be implemented from each customer's aggregated Total Sales: **Low Value** < ■100,000; **Medium Value** ■100,000–■249,999; **High Value** ■250,000–■399,999; **Very High Value** ≥ ■400,000. These thresholds are project-defined business rules, not statistically discovered cutoffs. The dashboard should display the resulting customer counts and revenue contribution after the segmentation is calculated.

10. Interactive Dashboard Design

Page 1 — Executive Overview: KPI cards for Revenue, Orders, Customers, Quantity and AOV; Revenue by Product; Revenue by City; Revenue by Age Group; Revenue by Gender; slicers for Product, City, Gender and Age Group.

Page 2 — Customer Deep Dive: Top 10 Customers, Customer Segment distribution, Revenue by Customer Segment, Customer revenue table, and drill-through from customer to transaction-level details.

Page 3 — Revenue Drivers: Quantity vs Total Sales scatter plot, Unit Price vs Total Sales scatter plot, Product × Age Group matrix, Product quantity vs revenue comparison, and City × Product analysis.

11. Recommended Business Actions

- Monitor high-revenue products, especially Laptop, when reviewing inventory and promotional priorities.
- Investigate why Mobile leads unit volume while Laptop leads revenue.
- Use city-level analysis to investigate the conditions associated with stronger revenue markets.
- Use customer-level spending and the defined segmentation framework for retention and engagement analysis.
- Review pricing and product mix because Unit Price has a relatively strong positive association with Total Sales.
- Avoid treating Age as a direct sales driver because its individual linear correlation with Total Sales is near zero.

12. Limitations

The analysis is based on the available cleaned sales dataset and is observational. Correlation does not establish causation. The customer segmentation thresholds are analyst-defined and should be reviewed with business stakeholders. A statistical hypothesis test and confidence interval should be calculated from the transaction-level dataset before reporting inferential conclusions.

13. Deliverable Checklist

- Deep-dive analysis report — included in this file.
- SQL analysis queries — supplied separately in business_analysis.sql.
- Power BI dashboard — build using Dashboard_Build_Guide.pdf.
- GitHub project README — supplied separately.
- 5–7 minute LinkedIn dashboard demonstration — slide storyboard supplied separately.

Source: ApexPlanet Task 2 EDA report and analysis results supplied in the project files.